

FOLLICULAR ROSE GARDEN

TRANSFORMING BURSTING

EXPLODING

WALDENSTRÖM'S MISTY POND

t(14;18)

BCL2

BR

WATCH

WAIT

3%/YR

POD24

3RD LINE

ENZYME INHIBITOR

MOSUN

TAMU

TARGET EZH2

SPLNECTOMY

HOVERS

HCV

DAA

CRUSHED INTO FIRST

NODAL

IgM FLARE

SPLING UPWARD

FILTERING

FILTERING

MARGINAL ZONE HERB PATCH

WALDENSTRÖM'S MISTY POND

>90% MYD88 L265P

THICK

11:18

ANTIBIOTICS

KILLING BACTERIA

MUCOSAL TISSUES

1. Follicular rose garden t(14;18)

WHAT YOU SEE

'(14;18)' banner over rose beds

WHAT IT ENCODES

Follicular lymphoma is the most common INDOLENT NHL (~20-22% of all NHL) and the second most common NHL overall after DLBCL. Its hallmark is t(14;18)(q32;q21), which juxtaposes the BCL2 gene on chromosome 18 with the IgH enhancer on chromosome 14, causing BCL2 overexpression. Histology shows a nodular/follicular growth pattern of small cleaved cells (centrocytes) and larger centroblasts; grading 1-3 is by centroblast count, and grade 3B behaves like an aggressive lymphoma. Mnemonic: Follicular = 'Fourteen-eighteen,' BCL2 the gardener.

BOARD TRAP

WRONG to attribute t(14;18)/BCL2 to Burkitt (that is MYC, t(8;14)) or to mantle cell (that is cyclin D1, t(11;14)). BCL2 overexpression by translocation is the follicular signature. Also, BCL2 protein positivity by IHC is nonspecific (also seen in normal T cells and other lymphomas) — it is the translocation, with a follicular architecture, that points to FL.

2. BCL2 watering can

WHAT YOU SEE

'BCL2' on watering can

WHAT IT ENCODES

BCL2 is an anti-apoptotic protein on the outer mitochondrial membrane that blocks programmed cell death. In follicular lymphoma, overexpressed BCL2 lets B cells survive far longer than they should, so the tumor accumulates SLOWLY rather than proliferating rapidly (low Ki-67, typically <20%). This 'failure to die' rather than 'drive to divide' biology explains why FL is indolent yet incurable — the cells persist and reseed despite therapy. Mnemonic: BCL2 waters the garden of immortal cells; nothing wilts.

BOARD TRAP

Confusing the indolent biology (defective apoptosis, low proliferation) with aggressive biology (high proliferation) is the trap. A follicular case with a HIGH Ki-67 or sheets of large cells suggests grade 3B or transformation to DLBCL, which changes management from observation toward aggressive chemoimmunotherapy.

3. WAIT bench

WHAT YOU SEE

Patient sitting on 'WAIT' bench

WHAT IT ENCODES

For asymptomatic, low-tumor-burden follicular lymphoma, the standard is watch-and-wait (active surveillance): randomized data show no survival benefit to starting therapy early versus deferring it until symptoms or high burden appear. Treatment triggers are the GELF criteria — bulky mass (>7 cm), ≥3 nodal sites each >3 cm, B symptoms, splenomegaly, cytopenias, effusions, or organ compromise. First treatment is typically rituximab ± chemo (bendamustine-R or R-CHOP) or, for very low burden, single-agent rituximab. Mnemonic: low-burden FL — sit on the WAIT bench.

BOARD TRAP

Treating asymptomatic low-burden follicular lymphoma with immediate chemotherapy is the classic trap — it adds toxicity without prolonging life. Observation is correct until a GELF/symptomatic trigger is met. (Note this logic is the opposite of aggressive lymphomas, where you treat urgently.)

4. 3%/yr transformation

WHAT YOU SEE

'3%/YR' near bursting/exploding tree

WHAT IT ENCODES

Follicular lymphoma transforms to an aggressive lymphoma (usually DLBCL) at a roughly constant rate of about 3% per year. Clues to transformation: rapid/asymmetric nodal growth, a sudden rise in LDH, new B symptoms, hypercalcemia, or a single site outpacing others. Transformation markedly worsens prognosis and demands re-biopsy of the most active/PET-avid site to confirm before switching to aggressive therapy (e.g., R-CHOP ± transplant/CAR-T). Mnemonic: FL leaks 3% per year into the aggressive pond.

BOARD TRAP

Continuing watch-and-wait when a follicular patient develops a rapidly enlarging node with rising LDH is the trap — that picture is transformation to DLBCL until biopsy proves otherwise, and it requires aggressive treatment, not observation.

5. Transforming tree

WHAT YOU SEE

Tree-archer 'TRANSFORMING / BURSTING'

WHAT IT ENCODES

Histologic transformation means the indolent follicular clone evolves (often via TP53 loss, MYC or additional hits) into a high-grade lymphoma, most commonly DLBCL — sometimes a 'double-hit' lymphoma if MYC plus BCL2/BCL6 rearrangements coexist. Always biopsy the most metabolically active lesion (highest SUV on PET) to capture the transformed component, since different sites may show different grades. Mnemonic: the calm tree suddenly bursts into an aggressive bloom — biopsy the brightest branch.

BOARD TRAP

Biopsying a convenient or low-activity node and missing the transformed clone is the sampling trap — you can under-stage the disease. Target the highest-SUV/most rapidly enlarging site to detect transformation.

6. 3rd-line van

WHAT YOU SEE

'3RD LINE' van, exploding

WHAT IT ENCODES

Relapsed/refractory follicular lymphoma (after multiple prior lines) has an expanding salvage menu: lenalidomide + rituximab (R2 regimen), EZH2 inhibitor (tazemetostat), PI3K inhibitors (historically), CD19 CAR-T (axicabtagene/tisagenlecleucel), and CD20xCD3 bispecific antibodies (mosunetuzumab). Early relapse within 24 months of first-line chemoimmunotherapy (POD24) identifies a high-risk subgroup with shortened survival. Mnemonic: when FL keeps coming back, escalate — R2, tazemetostat, CAR-T, bispecifics.

BOARD TRAP

Treating POD24 (progression within 24 months of frontline therapy) as routine relapse is the trap — POD24 is a powerful adverse prognostic marker that warrants re-biopsy to exclude transformation and consideration of more aggressive/clinical-trial therapy.

7. EZH2 inhibitor

WHAT YOU SEE

'ENZYME INHIBITOR / TARGET EZH2' antibody

WHAT IT ENCODES

Tazemetostat is an oral EZH2 inhibitor approved for relapsed/refractory follicular lymphoma. EZH2 is a histone methyltransferase (part of PRC2) that silences tumor-suppressor genes; activating EZH2 mutations occur in ~20% of FL, and tazemetostat is especially active in EZH2-MUTANT disease (also approved for EZH2-wildtype after ≥2 prior lines). Mnemonic: EZH2 'writes' repressive methyl marks — tazemetostat erases the advantage.

BOARD TRAP

Assuming tazemetostat works only in EZH2-mutant FL is a partial trap — it has activity (and approval) in EZH2-wildtype relapsed FL too, though response rates are higher in the mutant subgroup. Confirming EZH2 mutation status helps predict, but is not strictly required, for use.

8. MALT — killing bacteria

WHAT YOU SEE

'MUCOSAL TISSUES / KILLING BACTERIA' with t(11;18)

WHAT IT ENCODES

Extranodal marginal-zone lymphoma of mucosa-associated lymphoid tissue (MALT) is an indolent B-cell lymphoma arising at sites of chronic antigen stimulation. Gastric MALT is driven by chronic Helicobacter pylori infection; for early-stage, H. pylori-positive gastric MALT, H. pylori ERADICATION (triple/quadruple therapy) alone induces lymphoma regression in the majority of patients. Other MALT sites/triggers: ocular adnexa (Chlamydia psittaci), small bowel (Campylobacter jejuni), salivary gland (Sjögren), thyroid (Hashimoto). Mnemonic: kill the bug, cure the MALT.

BOARD TRAP

Treating t(11;18)-POSITIVE gastric MALT with antibiotics alone is the classic trap — t(11;18)+ tumors do NOT respond to H. pylori eradication and require radiation or systemic therapy (e.g., rituximab). Antibiotic therapy is reserved for early-stage, H. pylori-positive, t(11;18)-negative disease.

9. t(11;18) MALT

WHAT YOU SEE

'11;18' tag in MALT bed

WHAT IT ENCODES

t(11;18)(q21;q21) fuses API2-MALT1 and is the most common translocation in MALT lymphoma. It is a marker of more advanced/disseminated disease and, critically, predicts FAILURE of H. pylori eradication therapy — these tumors will not regress with antibiotics. Such patients are directed to radiotherapy (localized disease) or rituximab-based systemic therapy. Mnemonic: 'eleven-eighteen' = antibiotics in vain.

BOARD TRAP

WRONG to expect antibiotic-induced remission in a t(11;18)+ gastric MALT. The translocation flags antibiotic resistance and the need for radiation or immunochemotherapy; relying on H. pylori eradication delays effective treatment.

10. Splenic MZL

WHAT YOU SEE

'SPLENIC — DAH CRUSHED FIRST LINE'

WHAT IT ENCODES

Splenic marginal-zone lymphoma presents with massive splenomegaly, cytopenias, and circulating villous lymphocytes (B cells with short polar villi), usually WITHOUT lymphadenopathy. It is strongly associated with hepatitis C. Management: if HCV-positive, antiviral therapy first; if symptomatic and HCV-negative, rituximab (preferred) or splenectomy controls disease. Mnemonic: big spleen + villous lymphocytes + HCV = splenic MZL.

BOARD TRAP

Jumping to chemotherapy in an HCV-positive splenic MZL is the trap — antiviral (direct-acting) therapy alone can produce lymphoma regression and is first-line in HCV-driven disease. Reserve rituximab/splenectomy for HCV-negative or antiviral-refractory symptomatic cases.

11. HCV splenectomy link

WHAT YOU SEE

'HCV / SPLENECTOMY' label

WHAT IT ENCODES

Hepatitis C is a recognized driver of marginal-zone (especially splenic) lymphoma through chronic B-cell antigen stimulation; HCV is also the leading cause of mixed cryoglobulinemia. Eradicating HCV with direct-acting antivirals can cause the lymphoma to regress without cytotoxic therapy, which is why HCV testing is part of the lymphoma workup. Splenectomy is an alternative for symptomatic, antiviral-ineligible splenic MZL. Mnemonic: treat the virus, shrink the lymphoma.

BOARD TRAP

Forgetting to test for HCV in a new marginal-zone lymphoma is the workup trap — missing it means missing a curative antiviral approach. The same chronic-antigen logic explains why H. pylori (MALT) and HCV (splenic MZL) lymphomas can regress when the trigger is removed.

12. Nodal MZL

WHAT YOU SEE

The plain herb bed simply signposted NODAL in the marginal-zone herb patch — no bug, no spleen, no virus icon beside it, signaling nodal MZL is the driver-less, purely lymph-node marginal-zone subtype.

WHAT IT ENCODES

Nodal marginal-zone lymphoma is the rarest of the three marginal-zone subtypes (extranodal/MALT, splenic, nodal). It is a primary lymph-node disease without the extranodal or splenic pattern, is generally indolent, and is treated like other low-grade B-cell lymphomas (observation if asymptomatic; rituximab ± chemo when treatment is indicated). Mnemonic: three marginal zones — MALT (mucosa), splenic (spleen), nodal (nodes, rarest).

BOARD TRAP

Conflating the marginal-zone subtypes is the trap — only MALT/extranodal disease is antibiotic-responsive (H. pylori), splenic MZL ties to HCV, and nodal MZL has no infectious driver and is managed like a generic indolent nodal lymphoma.

13. MYD88 L265P — Waldenström

WHAT YOU SEE

Tombstone '>90% MYD88 L265P'

WHAT IT ENCODES

Waldenström macroglobulinemia is a lymphoplasmacytic lymphoma producing a monoclonal IgM paraprotein. The MYD88 L265P mutation is present in >90% of cases (a near-defining marker), and CXCR4 mutations modify presentation and ibrutinib response. The marrow shows lymphoplasmacytic cells; serum shows an IgM spike. Treatment options include rituximab-based regimens, bendamustine-R, and BTK inhibitors (ibrutinib/zanubrutinib). Mnemonic: Waldenström whispers 'L265P' and 'IgM.'

BOARD TRAP

Calling a monoclonal IgM gammopathy 'multiple myeloma' is the trap — myeloma is typically IgG/IgA with lytic bone lesions and rarely IgM, while an IgM monoclonal protein with lymphoplasmacytic marrow and MYD88 L265P is Waldenström. IgM does not usually cause the lytic bone disease/hypercalcemia of myeloma.

14. IgM hyperviscosity / flare

WHAT YOU SEE

'IgM FLARE — SPIKING UPWARD' bottles

WHAT IT ENCODES

The large pentameric IgM in Waldenström can raise serum viscosity, producing hyperviscosity syndrome: blurred vision/'sausage-link' retinal veins, headache, mucosal bleeding, and neurologic changes. Symptomatic hyperviscosity is a medical emergency treated with PLASMAPHERESIS (plasma exchange) for rapid IgM removal, then definitive lymphoma therapy. Rituximab can transiently INCREASE IgM ('IgM flare'), worsening viscosity. Mnemonic: thick blood — exchange first, then treat.

BOARD TRAP

Giving rituximab MONOTHERAPY when the IgM is very high (e.g., >40 g/L) is the trap — it can trigger an IgM flare and precipitate hyperviscosity. In that setting, lower the IgM first (plasmapheresis) and/or use a non-rituximab-first regimen; treat symptomatic hyperviscosity with plasma exchange, not transfusion (which can worsen viscosity).